

D²FL: Dimensional Disaster-oriented Backdoor Attack Defense Of Federated Learning

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Abstract—Defense algorithms for backdoor attacks in federated learning (FL) commonly rely on model parameter vectorization. However, as neural networks deepen, the exponential growth of model parameters leads to increased dimensionality, exacerbating the curse of dimensionality and reducing the effectiveness of traditional distance-based defenses. To address this, we propose Dimensional Disaster-oriented Backdoor Attack Defense in Federated Learning (D²FL), a method that mitigates attacks by focusing on expressive backdoor modules rather than the entire model. This approach reduces dimensionality and mitigates the challenges posed by large parameter spaces. Our extensive evaluation of D²FL on image classification tasks across various deep neural networks demonstrates its superior efficiency, significantly reducing both defense and aggregation times.

Index Terms—Federated Learning, Backdoor Attack Defense, Dimensionality Disaster.

I. INTRODUCTION

Federated learning (FL) is a distributed machine learning algorithm used in fields like mobile keyword prediction [1]. Clients in FL train local models using their data and upload the updated parameters to a central server. The server aggregates these parameters into a global model, which is then distributed back to clients. While FL protects privacy through decentralized training, its distributed nature introduces vulnerabilities. Attackers can exploit these vulnerabilities by performing malicious operations locally and submitting harmful updates.

Deficiencies of existing defenses. Existing defenses against backdoor attacks involve four phases: feature extraction, model differentiation, model aggregation, and update restriction [2]–[6]. In feature extraction, Euclidean distance [2] or cosine similarity [3] is used to represent model parameters. In model differentiation, strategies such as clustering, the ReLU function [4], or other algorithms calculate user model parameter weights [2]. The third phase aggregates the global model using these weights, while the fourth phase applies update restrictions through noise addition or step size limits.

Feature extraction often involves vectorizing model updates to simplify feature calculation. As neural networks deepen, their parameters increase, shifting data from low to high dimensional spaces. This shift increases the likelihood of encountering the curse of dimensionality, reducing algorithm effectiveness. In high dimensions, distance functions lose significance due to sparsity, and clustering algorithms become

less effective [7], [8]. Thus, distance-based defenses fail to identify malicious parameters, allowing poisoned updates in global model aggregation. Vectorizing all model parameters increases the computational burden, especially in high-concurrency federated learning, decreasing efficiency. Reducing dimensionality may mitigate these issues and enhance defense performance.

We propose a new method to address the curse of dimensionality and efficiency bottlenecks in feature extraction. We introduce a cross-swapping method to identify backdoor information and reduce the dimensionality of neural network parameters. This minimizes the curse of dimensionality and improves defense efficiency and performance through enhanced backdoor detection.

Our Goals and Contributions. We introduce D²FL, a defense strategy for mitigating backdoor attacks in federated learning, particularly in high-dimensional scenarios with large, deep models. D²FL improves defense performance with a modified approach, preserving the aggregated model's effectiveness. It is compatible with general defense algorithms that rely on model parameter vectorization, without assuming a specific adversarial strategy or data distribution for benign and adversarial datasets. Our contributions are as follows:

- We introduce Dimensional Disaster-oriented Backdoor Attack Defense (D²FL), a novel defense method for mitigating backdoor attacks in federated learning. D²FL addresses the curse of dimensionality in deep neural network models while significantly improving the performance and efficiency of the original defense algorithm.
- To the best of our knowledge, we are the first to propose an algorithm that addresses the curse of dimensionality in federated learning defense algorithms for high-dimensional data scenarios.
- Our experiment shows that the classifier's parameters are critical in determining whether the final output contains backdoor information. This conclusion is based on cross-swapping parameters between the backdoor and clean models during inference.

II. BACKGROUND AND RELATED WORK

A. Federated Learning

FL [9] is a distributed framework with n clients and a central server. As shown in Figure 1, it involves parallel training of

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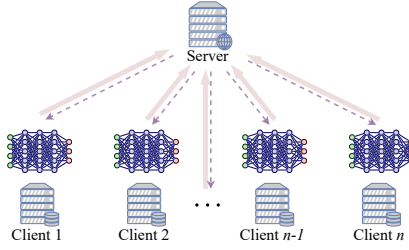


Fig. 1: The basic process of federated learning is illustrated, with the central server at the top of the diagram and the clients represented by the modules at the bottom. The connections in the middle depict the distribution and upload of model parameter updates between the central server and the clients.

client models, which are aggregated into a global model G . In each training iteration $t \in 1, \dots, T$, client $i \in 1, \dots, n$ locally trains a model w_i with p parameters (weights and biases) using local data D_i based on the previous global model G_{t-1} . The locally trained models w_i are sent to the aggregator, which updates the global model G_t . In FedAvg, the global model is generated by averaging the model parameters from all clients: $G_t = \sum_{i=1}^n w_i / n$. This approach assumes equal weighting for each client's model, i.e., $1/n$. Several aggregation algorithms have been proposed, including FedAvg [9], Krum [2], and Median [10]. However, many defense algorithms, such as Flame [3], FLTrust [4], DeepSight [6], and Rflbat [5], use FedAvg as the aggregation mechanism. Since FedAvg is widely used in FL and backdoor attack studies, we focus on it in this study. In this study, we consider this modified weighting approach as part of FedAvg.

B. Backdoor Attacks on Federated Learning

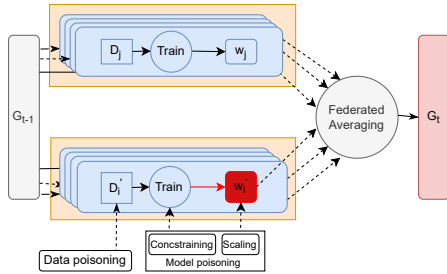


Fig. 2: An overview of backdoor attacks

In backdoor attacks, one or more adversaries manipulate k controlled clients' local models w_i to create poisoned models w_i' , which are then aggregated into the global model G_t . This implants a backdoor into the global model, allowing it to perform malicious tasks. The adversary ensures that the poisoned model G_t' maintains performance on the main task, similar to a non-poisoned model, preserving accuracy on normal data. Figure 2 illustrates common techniques in FL backdoor attacks: 1) data poisoning, e.g., [11], where the

adversary manipulates the dataset, and 2) model poisoning, e.g., [12], where the adversary alters the training process or model. These attack techniques are briefly discussed below.

Data Poisoning. The adversary alters the local data of compromised clients, turning benign data into poisoned data, and then trains a backdoored model. The adversary submits the poisoned model parameters to the server. These alterations include flipping labels, embedding pixel pattern triggers [13], or modifying image colors [14]. The malicious client ratio (mr) is the proportion of malicious clients controlled by the adversary in each federated round, calculated as $mr = m/N$, where m is the number of malicious clients and N is the total number of clients.

Model Poisoning. This attack requires the adversary to control multiple clients. The adversary modifies the clients' training data and alters the model training process by adjusting parameters and scaling updates. This maximizes the attack's impact while avoiding detection by the central server. Methods include: 1) enlarging weights to increase impact [15], 2) scaling updates to evade detection [15], and 3) constraining training to minimize differences between models and reduce detection likelihood [15].

Backdoor attacks using data poisoning require only the adversary's ability to modify data from controlled clients, while federated learning's black-box procedures reduce the privileges needed compared to model poisoning. Data poisoning-based backdoor attacks are more prevalent than those using model poisoning. This study focuses on investigating these attacks.

The Curse of Dimensionality is a well-known issue in high-dimensional spaces, especially in data science, statistics, and machine learning. It refers to challenges that grow significantly as data dimensionality increases. The manifestations of the curse of dimensionality include: 1) Data Sparsity: As dimensionality increases, data points become increasingly sparse, making statistical analysis and modeling challenging, as most points are far apart in high-dimensional space. 2) Computational Complexity: High-dimensional data requires significantly more computational resources. The time complexity of many algorithms increases exponentially with dimensionality, making computations impractical. 3) Distance Metric Breakdown: In high-dimensional space, distances between data points lose their distinguishing power. Common distance metrics may become ineffective, hindering many distance-based algorithms.

After vectorization, model parameters are mapped to a high-dimensional space, becoming sparse in this space. As a result, traditional defense algorithms face higher computational demands and are less effective due to data sparsity. Both distance-based and clustering algorithms face challenges under these conditions.

III. THREAT MODEL

In this work, we focus on data poisoning backdoor attacks, which do not require control or knowledge of the training process, including model architecture, parameters, or training

methods. These attacks pose a significant threat in real-world scenarios as they do not necessitate any computational resources from clients or central servers [16]. Given the prevalence of image classification tasks in backdoor attack studies, we adopt them as the primary focus of our research.

Our study assumes a fixed-frequency attack setting [16] and evaluates our approach using the commonly utilized BadNet trigger. Specifically, for the MNIST dataset, we employ a white square marking four pixels in the upper right corner as the trigger. For the CIFAR-10 dataset, we use a black cross marking five pixels in the upper right corner, with a uniform backdoor label set to 0. Examples of backdoor triggers are shown in Figure 3.

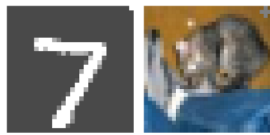


Fig. 3: The backdoor triggers. For the MNIST dataset (left), the trigger is a white square marking four pixels in the upper right corner, while for the CIFAR-10 dataset (right), the trigger is a black cross marking five pixels in the upper right corner.

IV. STRATEGY OVERVIEW AND DESIGN

A. High-level Idea

As deep learning evolves, neural network models are becoming deeper and larger. Current defenses against backdoor attacks involve four steps: feature extraction, model discrimination, aggregation, and update restriction. In feature extraction, metrics like Euclidean distance [2] or cosine similarity [3] are used to represent model parameters. In model discrimination, strategies such as clustering, ReLU [4], or other algorithms compute weights for model parameters. The third step aggregates the global model using these weights. The fourth step adds noise or restricts the update step size to obtain a constrained global model. Not all defense algorithms follow these steps, but they all require feature extraction, usually by vectorizing model parameters. Defense algorithms that vectorize model parameters face challenges when transforming them into high-dimensional vectors, causing the curse of dimensionality. This issue diminishes defense performance on larger, deeper models. As model parameters increase with greater depth, inspection time grows, especially in high-concurrency scenarios, worsening inefficiencies. Reducing the dimensionality of model parameters in deep neural networks mitigates catastrophic failures in defense algorithms, improving performance and efficiency.

B. Neural Networks Crossover Design

We designed an experiment inspired by chromosome crossover [17] to investigate the representation of backdoor information across neural network modules. The procedure, outlined in Algorithm 1, involves training two identical networks on clean and poisoned datasets, yielding a clean model

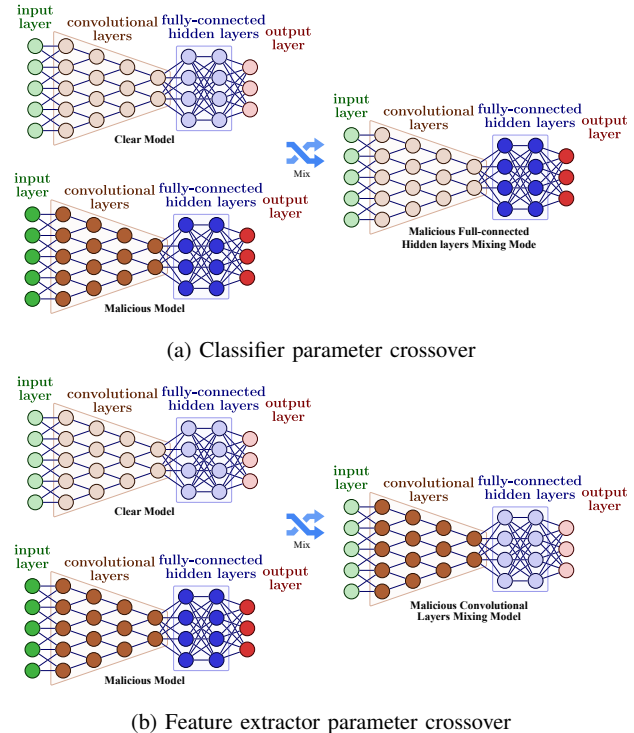


Fig. 4: Neural Network Crossover. In the neural network structure, the parameters of the classifier (a) and the feature extractor (b) are crossover separately.

and a backdoored model. After testing the main task and backdoor task accuracies for both models, we perform module crossover. To create a hybrid model with a feature extractor containing backdoor information and a classifier without it, we combine the classifier from the clean network with the feature extractor from the backdoored network, resulting in the model classifier. Conversely, to create a hybrid model with a classifier containing backdoor information and a clean feature extractor, we combine the feature extractor from the backdoored network with the classifier from the clean network, resulting in the model feature. Figure 4 illustrates the process of generating these hybrid models. We then compare the main and backdoor task accuracies of the model-classifier and model-feature to determine the representation of backdoor information in the classifier and feature extractor.

C. Strategy Design

In federated learning backdoor attack defenses, techniques such as weight selection (Median) [10], Euclidean distance calculation (Krum, Mul-Krum, Rflbat) [2], [5], and cosine similarity for clustering (Flame, Deepsight, Fltrust) [3], [4], [17] are commonly used. However, as model parameters grow exponentially, particularly in large language models, vectorizing these parameters for inspection or clustering incurs substantial computational overhead and exacerbates the curse of dimensionality [18]. Traditional clustering algorithms struggle

Algorithm 1: Neural Network Architecture Crossover

Input: Clean dataset: *ClearData*, Backdoor dataset: *BackdoorData*

```
1  $i \leftarrow 0$ ;
2 for  $\text{traindata} \leftarrow \text{ClearData}$  to  $\text{BackdoorData}$  do
3    $\text{Model}[i] \leftarrow \text{Train}(\text{train\_data})$ ;
4    $i \leftarrow i + 1$ ;
5 end
6 for  $\text{layer} \leftarrow \text{classifier}$  to  $\text{feature}$  do
7    $\text{Model}[i] \leftarrow \text{Combine}(\text{Model}[0], \text{Model}[1], \text{layer})$ ;
8    $i \leftarrow i + 1$ ;
9 end
10 for  $\text{model} \leftarrow \text{Model}[3]$  to  $\text{Model}[0]$  do
11    $\text{MA}[i] \leftarrow \text{Evaluate}(\text{model}, \text{ClearData})$ ;
12    $\text{BA}[i] \leftarrow \text{Evaluate}(\text{model}, \text{BackdoorData})$ ;
13    $i \leftarrow i - 1$ ;
14 end
```

in high-dimensional spaces, posing challenges for backdoor defense. To address this, defenses must reduce the number of parameters inspected or the dimensionality post-vectorization. In our neural network crossover experiment, combining a clean feature extractor with a backdoored classifier produced a hybrid model with strong backdoor capabilities, while combining a clean classifier with a backdoored feature extractor yielded a model with strong clean task performance but weak backdoor capabilities. Neural networks exhibit pronounced backdoor behavior in classifiers trained with backdoor data, and leveraging this can help mitigate dimensionality challenges in defense algorithms. We propose D²FL, a defense strategy against backdoor attacks in federated learning, as outlined in Algorithm 2.

Algorithm 2: Dimensional Disaster-oriented Backdoor Attack Defense

Input: client model $w_i, \dots, w_\tau$

Output: global model w_g

```
1 for  $i \leftarrow 0$  to  $\tau$  do
2    $w_i^{fc} \leftarrow \text{extract}(w_i)$ 
3  $\text{good}_i, \dots, \text{good}_n \leftarrow \text{recognize}(w_0^{fc}, \dots, w_\tau^{fc})$ 
4  $w_g = \text{aggregate}(w_{\text{good}_i}, \dots, w_{\text{good}_n})$ 
5 return  $w_g$ 
```

Our method extracts classifier parameters, specifically from the fully connected layer, rather than vectorizing the entire model. This reduces dimensionality, mitigates the risk of dimensionality disaster, and preserves potential backdoor information, thus lowering computational cost. To evaluate the strategy’s robustness, we propose two defense algorithms—Small Flame and Small FLTrust—derived from the Flame and FLTrust algorithms, respectively. These are tested on the CIFAR10 dataset [19] with LeNet [20], ResNet18 [21], and ResNet101 [21], followed by trigger attacks after fifty learning rounds.

V. EXPERIMENTAL SETUP

In this study, image classification tasks were selected due to their widespread real-world applications and common use

in backdoor attack research. The experiment involved 100 clients, as in [4], [22], with 15% randomly selected to submit model parameters per federated learning round. The model replication rate (mr) was set to 30%. To simulate the initial phase of federated learning without attacks, no attacks were introduced during the first fifty training rounds. To model client data heterogeneity, we used a Dirichlet distribution with $\alpha = 0.5$ [22], representing non-independent and identically distributed (non-IID) data. Backdoor attacks were introduced from the fiftieth round onward, using the FedAvg algorithm as the baseline for aggregation.

Experimental Platform: The experiment was conducted on Ubuntu 20.04.5 LTS, with an AMD EPYC 9754 (256 cores, 2.25 GHz), 64GB memory, and an NVIDIA GeForce RTX 4090D GPU. The system used CUDA 11.8 and PyTorch 2.0.0. We open-source our code in github: <https://github.com/BESTICSP/D2FL>.

We evaluate the effectiveness of backdoor attack and defense techniques using the following metrics: 1) Main Task Accuracy (MA), which measures the accuracy of a model on its primary task, reflecting the fraction of benign inputs correctly classified. The adversary seeks to minimize its impact to avoid detection. 2) Backdoor Accuracy (BA), which evaluates performance on the backdoor task by counting correct backdoor predictions. Attackers aim to maximize this metric. 3) Main Task Accuracy Average Improvement (MAI) quantifies defense effectiveness by calculating the average increase in MA per round compared to the baseline defense (FedAvg), as shown in Eq1. 4) Backdoor Accuracy Average Reduction (BAR) measures the defense’s ability to reduce backdoor task accuracy relative to the baseline defense, as shown in Eq2. Time Average Improvement (TAI) assesses the efficiency of a defense algorithm by comparing its training time to that of the baseline, as shown in Eq3.

$$\text{MAI} = (\sum_{i=1}^n \text{ma}_i^{\text{defense}} - \text{ma}_i^{\text{baseline}}) / n \quad (1)$$

$$\text{BAR} = (\sum_{i=1}^n \text{ba}_i^{\text{baseline}} - \text{ba}_i^{\text{defense}}) / n \quad (2)$$

$$\text{TAI} = (\sum_{i=1}^n \text{time}_i^{\text{defense}} - \text{time}_i^{\text{baseline}}) / n \quad (3)$$

VI. EXPERIMENTAL RESULTS

This study presents two main experimental results. First, we detect the main task and backdoor task accuracy after neural network module crossover. Second, we compare the main task accuracy, backdoor task accuracy, and computation time between the improved Flame and FLTrust strategies and the original versions, using the same neural network architectures with varying depths. The first result assesses the expression of backdoor information in the main and backdoor tasks of the implanted networks, while the second tests the practical effectiveness of the strategies.

A. Crossover Comparison

By using the method of neural network module crossover, CNN, LeNet, MobileNetV2 [23], ResNet18, and VGG13 [24]

were trained as normal models and with trigger backdoor information implanted. The neural networks were then disassembled and reassembled through module crossover to obtain different sub-models. Finally, the main task accuracy and backdoor task accuracy of all models were tested.

Models are classified based on the presence of backdoor information in their parameters as follows: 1) Parental Models: CNN, LeNet, MobileNetV2, ResNet18, and VGG13 models trained on clean data. 2) Maternal Models: CNN, LeNet, MobileNetV2, ResNet18, and VGG13 models trained on backdoor data. 3) Filial 1 Models: Models created by combining the parental model's feature extractor with the maternal model's classifier. 4) Filial 2 Models: Models created by combining the maternal model's feature extractor with the parental model's classifier. Parental and maternal models serve as baselines.

To enhance the comparative analysis, we trained two sets of highly differentiated parental models. From Table I, the paternal model set achieved high performance on the main task with minimal backdoor task capability, while the maternal model set excelled in the backdoor task but underperformed on the main task. Due to parameter crossover, both Main Task Accuracy (MA) and Backdoor Task Accuracy (BA) were impacted. Impact on MA: Filial 1 models, across all five architectures (CNN, LeNet, MobileNetV2, ResNet18, and VGG13), consistently showed significantly reduced performance on the main task. Filial 2 models also failed to match the performance of the parental models in the main task. Impact on BA: Filial 1 models achieved backdoor task performance comparable to the maternal models, whereas Filial 2 models underperformed on the backdoor task. Performance details for the Filial models on both tasks are provided in Table II.

TABLE I: The performance of the parental models in terms of accuracy on the main task and the backdoor task, respectively.

	Models	CNN	LeNet	MobileNetv2	ResNet18	VGG13
MA	Paternal	99.03%	98.96%	88.5%	81.14%	89.20%
	Maternal	9.80%	9.80%	10.00%	10.00%	10.00%
BA	Paternal	9.77%	10.45%	0.00%	0.00%	7.91%
	Maternal	100.00%	100.00%	100.0%	100.00%	100.00%

TABLE II: The performance of the sub-models in terms of accuracy on the main task and the backdoor task, respectively.

	Models	CNN	LeNet	MobileNetv2	ResNet18	VGG13
MA	Filial 1	9.80%	9.80%	10.00%	10.00%	10.00%
	Filial 2	7.17%	10.36%	10.00%	10.00%	10.00%
BA	Filial 1	100.0%	100.00%	100.0%	100.00%	100.00%
	Filial 2	0.00%	0.00%	0.00%	0.00%	0.00%

B. Overall Strategy Performance

Previous experiments show that data-side attacks infect a model's classifier with backdoor information, allowing it to achieve backdoor task accuracy similar to the parental model. In federated learning, as neural networks deepen, traditional defense methods based on model vectorization are vulnerable

to dimensionality disaster. To mitigate this, we propose a defense strategy that exploits the classifier's backdoor characteristic to address the dimensionality challenge in vectorization-based defenses.

1) *Performance aspects*: The primary goal of federated learning is to train a global model with high accuracy on the main task while preserving user privacy. In image classification, distinguishing between categories is essential. However, in the presence of backdoor attacks, defense algorithms must identify and mitigate malicious model parameters submitted by adversarial users. Thus, maintaining high main task accuracy is a critical metric for evaluating defense performance, while minimizing backdoor task accuracy is the goal for defense algorithms.

This paper improves defense strategies against backdoor attacks in federated learning, focusing on model parameter vectors. We propose two enhanced defense algorithms, Small Flame and Small FLTrust, which address the dimensionality disaster. Using the FedAvg algorithm as a baseline, we simulate non-IID data partitioning with the CIFAR10 dataset and train models on LeNet, ResNet18, and ResNet101. Figures 5 and 6 show the performance of Flame and Small Flame, as well as FLTrust and Small FLTrust, over 200 training rounds. Experimental results are summarized in Table III.

From Table III, we observe that for the shallow LeNet network, Small Flame's MAI improvement is similar to that of Flame. However, Small FLTrust's MAI is more than double that of FLTrust. For deeper networks, Small Flame slightly outperforms Flame on ResNet18, while Small FLTrust shows lower MAI than FLTrust. On ResNet101, both Small Flame and Small FLTrust outperform their original counterparts in MAI. In BAR, for LeNet, Small Flame with the enhanced strategy shows a slight advantage over Flame. Small FLTrust exhibits significantly higher BAR than FLTrust. On ResNet18, Small Flame shows a higher BAR compared to the original strategy, while Small FLTrust has a slightly lower BAR than FLTrust. On ResNet101, both Small Flame and Small FLTrust demonstrate superior average BAR compared to their original strategies.

TABLE III: The performance of Flame, FLTrust under the original strategy, and Small Flame, Small FLTrust under the modified strategy across three different depths of neural networks: LeNet, ResNet18, and ResNet101.

Defenses	LeNet		ResNet18		ResNet101	
	MAI	BAR	MAI	BAR	MAI	BAR
Flame	18.75%	36.71%	17.41%	9.22%	23.94%	26.42%
Small Flame	18.36%	38.27%	20.51%	14.38%	35.78%	43.33%
FLtrust	7.68%	20.38%	28.08%	23.35%	28.30%	17.40%
Small FLtrust	20.36%	37.41%	26.31%	21.64%	32.94%	18.11%

2) *Aggregation time*: To evaluate the time efficiency of the defense algorithms, we use the MAI defined in Equation 3. Figure 7 shows the time required for aggregation after each training round for neural networks of varying depths, comparing the original and lightweight strategy improvements. Table IV

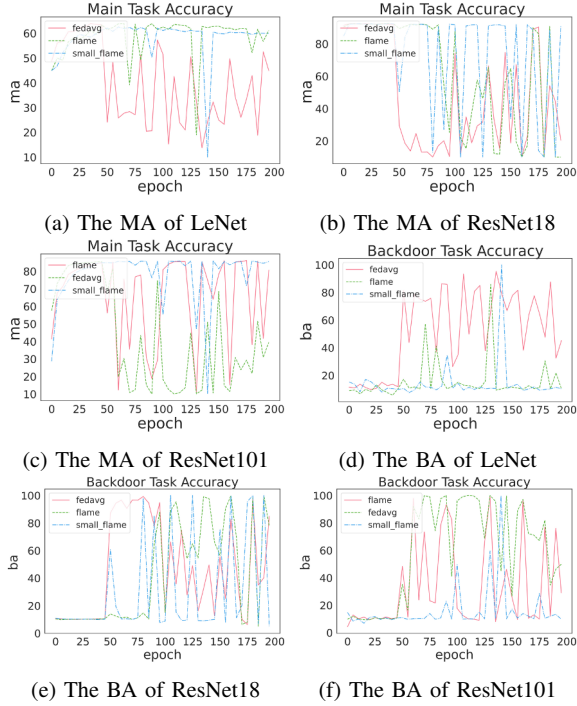


Fig. 5: Each column respectively shows the efficiency of Flame defense under different strategies for models of varying depths: LeNet, ResNet18, and ResNet101 over 200 training rounds.

presents the Time Mean Increase for different strategies across neural networks of varying depths.

TABLE IV: The TAI of Flame, Small Flame, FLTrust, Small FLTrust across LeNet, ResNet18, and ResNet101 models

Defenses	LeNet TAI	ResNet18 TAI	ResNet101 TAI
Flame	0.1248s	0.8019s	4.4573s
Small Flame	0.1301s	0.3084s	2.3489s
Fltrust	1.3452s	2.6836s	3.7969s
Small Fltrust	0.3650s	1.2984s	2.9326s

As shown in Figure 7, except for the similar time consumption of Flame and Small Flame on LeNet, lightweight strategies generally outperform the original strategies in time efficiency. Table IV indicates that, on the shallow LeNet network, FLTrust with the lightweight strategy significantly reduces TAI compared to the original. In ResNet18, the new strategy outperforms the original Flame and FLTrust strategies by a large margin. For the deeper ResNet101, the Small Flame algorithm maintains nearly double the time advantage, while Small FLTrust, though not doubling, still exhibits significantly lower TAI than the original strategy.

VII. DISCUSSION

As neural networks deepen, the vectorized model parameters grow increasingly high-dimensional, exacerbating the curse of dimensionality for defense algorithms. Our experiments reveal that the classifier module plays a crucial role

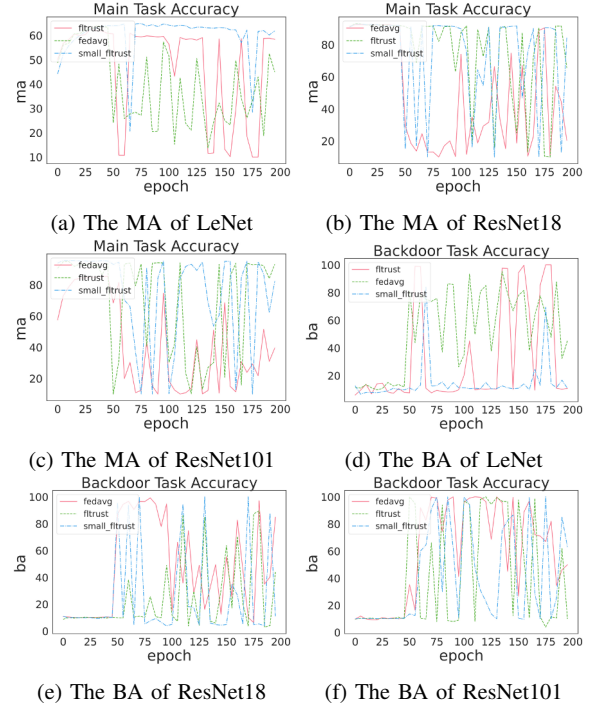


Fig. 6: Each column respectively shows the efficiency of FLTrust defense under different strategies for models of varying depths: LeNet, ResNet18, and ResNet101 over 200 training rounds.

in determining whether backdoor information appears in the model's output. Therefore, we propose using only the classifier parameters, rather than the full model parameters, to reduce dimensionality and mitigate the curse of dimensionality in defense algorithms. To evaluate this approach, we conducted experiments on neural networks of varying depths.

Our results show that D²FL significantly improves efficiency and accuracy across different network depths (see TableIII,IV). While designed to address the curse of dimensionality in deeper networks, D²FL also maintains or improves performance in shallower networks, such as LeNet. However, for defense algorithms that do not rely on distance or clustering functions, D²FL enhances computation speed without necessarily improving accuracy.

VIII. CONCLUSION

We propose a scheme for identifying backdoor representations in neural networks through module crossover, along with a novel federated learning defense method to address the curse of dimensionality. The experiments results show that classifier parameters in backdoored neural networks significantly express backdoor information. The key to mitigating the curse of dimensionality in federated learning defenses lies in using classifier parameters rather than full model parameters. We introduce three new evaluation metrics to assess defense performance in terms of main task accuracy, backdoor task

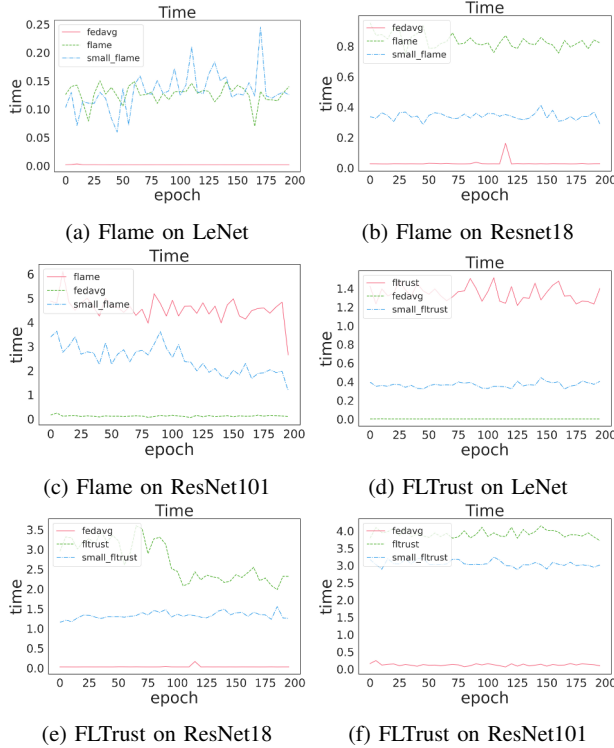


Fig. 7: Time consumption of Flame, Small Flame, FLTrust, Small FLTrust, and FedAvg on LeNet, ResNet18, and ResNet101 over 200 training rounds.

accuracy, and aggregation time compared to the FedAvg algorithm.

Our experiments confirm that backdoor information is strongly expressed in the classifiers of poisoned neural networks. Testing on models of varying depths shows that the new strategy outperforms the original as depth increases. Overall, the new strategy proves effective in improving main task accuracy, reducing backdoor task accuracy, and shortening aggregation time during federated learning. Next steps for practitioners: Our findings suggest that performing security checks on key model parameters, rather than all parameters, can enhance the efficiency of the checks. Practitioners should consider applying this strategy when deploying FL models. D²FL not only maintains prediction accuracy under non-IID settings but also reduces the backdoor rate, making it a promising option for practical applications.

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